

Manufacturing Downtime Analysis - Stakeholder Analysis

Executive Overview

This stakeholder analysis identifies all parties with an interest in the Manufacturing Downtime Analysis dashboard project, evaluates their influence and requirements, and establishes an engagement strategy to ensure project success.

1. Stakeholder Identification

Primary Stakeholders

1.1 Production Manager

Role: Executive Sponsor

Department: Manufacturing Operations

Influence Level: Very High

Interest Level: Very High

Needs & Expectations:

- Real-time visibility into production line performance
- Quick identification of downtime root causes
- Data to support operational improvement initiatives
- Weekly executive summary reports
- Mobile access for on-the-go monitoring

Pain Points:

- Currently relies on manual Excel reports with 24-48 hour delays
- Difficulty identifying patterns across multiple production lines
- Time-consuming data aggregation from various sources
- Limited visibility into operator-level performance

Success Criteria:

- Reduce time spent on manual reporting by 70%
- Enable same-day identification of downtime issues
- Support data-driven production planning decisions

1.2 Plant Floor Supervisors (4 persons)

Role: Primary End Users

Department: Production Floor

Influence Level: High

Interest Level: Very High

Needs & Expectations:

- Easy-to-use interface requiring minimal training
- Ability to filter by operator and product
- Clear visualization of Top 5 downtime factors
- Comparison capabilities between shifts
- Quick daily performance checks

Pain Points:

- Current manual logs are error-prone
- Difficult to track operator performance trends
- No standardized way to prioritize improvement efforts
- Time lag in receiving production reports

Success Criteria:

- Complete daily performance review in < 10 minutes
- Identify improvement opportunities independently
- Reduce downtime incidents by 15% within 3 months

Engagement Strategy:

- Include in weekly demo sessions
- Gather feedback through structured surveys
- Provide hands-on training sessions
- Designate as super-users for peer support

1.3 Quality Assurance Manager

Role: Key User & Data Validator

Department: Quality Assurance

Influence Level: Medium

Interest Level: High

Needs & Expectations:

- Correlation analysis between downtime and quality metrics
- Historical trend data for process improvement

- Variance analysis capabilities
- Export functionality for detailed analysis

Pain Points:

- Limited integration between production and quality data
- Manual effort to identify quality-downtime correlations
- Insufficient historical data accessibility

Success Criteria:

- Ability to identify quality-related downtime factors
- Access to 6+ months of historical data
- Export capability for statistical analysis

Engagement Strategy:

- Monthly review meetings
- Early access to beta features
- Collaboration on metric definitions

1.4 Production Operators (12 persons)

Role: Data Contributors & Indirect Users

Department: Production Floor

Influence Level: Low

Interest Level: Medium

Needs & Expectations:

- Fair and transparent performance tracking
- Clear understanding of how their data is used
- Recognition for high performance
- Actionable feedback on improvement areas

Pain Points:

- Concern about performance monitoring
- Uncertainty about data accuracy
- Limited visibility into their own metrics

Success Criteria:

- Trust in data accuracy and fairness
- Understanding of how dashboard supports them
- Positive impact on performance evaluations

Engagement Strategy:

- Transparent communication about dashboard purpose
- Anonymous performance summaries
- Regular feedback sessions
- Emphasize continuous improvement, not blame

Secondary Stakeholders**2.1 IT Department**

Role: Technical Support & Infrastructure

Influence Level: High

Interest Level: Medium

Responsibilities:

- Power BI licensing and access management
- Data source connectivity and security
- Technical troubleshooting
- Future cloud deployment

Requirements:

- Adherence to IT security policies
- Documentation of data connections
- Change management procedures
- Disaster recovery planning

Engagement Strategy:

- Early involvement in architecture decisions
- Regular technical sync meetings
- Provide comprehensive technical documentation
- Coordinate on deployment and maintenance

2.2 Finance Department

Role: Budget Approver & ROI Analyst

Influence Level: Medium

Interest Level: Low

Needs & Expectations:

- Clear ROI justification

- Budget tracking for tool licenses
- Cost-benefit analysis
- Quantifiable productivity improvements

Engagement Strategy:

- Provide ROI calculations and projections
- Quarterly business value reports
- Link improvements to cost savings

2.3 HR Department

Role: Performance Management Liaison

Influence Level: Low

Interest Level: Low

Concerns:

- Ethical use of operator performance data
- Privacy and data protection compliance
- Fair performance evaluation practices

Engagement Strategy:

- Ensure compliance with privacy policies
- Transparent communication about data usage
- Involve in operator communication planning

2. Power-Interest Matrix

High Power, High Interest (Manage Closely)

- Production Manager
- Plant Floor Supervisors

Strategy: Close collaboration, frequent updates, involve in key decisions

High Power, Low Interest (Keep Satisfied)

- IT Department
- Finance Department

Strategy: Regular updates, ensure requirements are met, minimal burden

Low Power, High Interest (Keep Informed)

- Quality Assurance Manager
- Production Operators

Strategy: Adequate information flow, engagement opportunities, address concerns

Low Power, Low Interest (Monitor)

- HR Department

Strategy: General updates, minimal engagement unless issues arise

3. Communication Strategy

Communication Channels

Stakeholder Group	Method	Frequency	Content Type
Production Manager	Email + In-person	Weekly	Status reports, key decisions
Plant Supervisors	Workshops + Email	Bi-weekly	Demos, training, feedback sessions
QA Manager	Email	Monthly	Feature updates, data quality
Production Operators	Team meetings	As needed	Purpose, benefits, Q&A
IT Department	Email + Technical docs	Bi-weekly	Technical requirements, issues
Finance	Email	Quarterly	ROI reports, budget updates
HR Department	Email	As needed	Policy compliance, concerns

4. Stakeholder Requirements Matrix

Functional Requirements

Requirement	Priority	Stakeholder(s)	Status
Product filtering	High	Supervisors, Production Mgr	✔ Implemented
Operator filtering	High	Supervisors, Production Mgr	✔ Implemented
Date range selection	High	All users	✔ Implemented
Top 5 downtime factors	High	Supervisors, QA Mgr	✔ Implemented
Trend analysis	Medium	Production Mgr, QA Mgr	✔ Implemented
Product × Operator matrix	Medium	Supervisors	✔ Implemented
Downtime % KPI	High	Production Mgr	✔ Implemented
Export to Excel	Medium	QA Mgr	□ Future phase

Requirement	Priority	Stakeholder(s)	Status
Mobile optimization	Low	Production Mgr	▢ Planned Phase 2
Batch detail page	Medium	Supervisors	▢ Planned Phase 2

Non-Functional Requirements

Requirement	Priority	Stakeholder(s)	Target
Dashboard load time	High	All users	< 5 seconds
Data refresh frequency	High	Production Mgr	Manual on-demand
User training duration	Medium	Supervisors	< 2 hours
System uptime	High	IT, Production Mgr	99.5%
Data accuracy	Very High	All users	100%

5. Resistance Management

Anticipated Resistance

Source: Production Operators

Reason: Fear of increased monitoring and job security concerns

Impact: Medium

Mitigation:

- Emphasize dashboard is for process improvement, not individual blame
- Transparent communication about data usage
- Anonymous reporting where possible
- Highlight examples of positive outcomes
- Involve operator representatives in feedback sessions

Source: Plant Supervisors

Reason: Change resistance, learning curve, time investment

Impact: Medium

Mitigation:

- Demonstrate time savings and efficiency gains
- Provide comprehensive training and support
- Quick wins to build confidence
- Peer champions to advocate for adoption

Source: IT Department

Reason: Concerns about support burden and security

Impact: Low

Mitigation:

- Thorough documentation
- Clear support procedures
- Adherence to IT policies
- Regular technical sync meetings

6. Engagement Plan

Phase 1: Requirements Gathering (Completed)

- ✓ One-on-one interviews with Production Manager
- ✓ Focus group with Plant Supervisors
- ✓ Survey of Production Operators
- ✓ Technical consultation with IT

Phase 2: Design & Development

- Weekly demos with Production Manager
- Bi-weekly feedback sessions with Supervisors
- Design review with QA Manager
- Technical checkpoints with IT

Phase 3: Testing & Refinement

- User Acceptance Testing with Supervisors
- Data validation with QA Manager
- Security review with IT
- Operator preview sessions

Phase 4: Deployment & Training

- Train-the-trainer session with Supervisors
- Group training for all users
- One-on-one support for Production Manager
- IT handoff and documentation

Phase 5: Post-Launch Support

- 30-day hyper-care with daily check-ins
- Weekly office hours for questions
- Monthly stakeholder feedback surveys
- Quarterly business review with sponsors

7. Success Metrics by Stakeholder

Production Manager

- **Time Savings:** 70% reduction in reporting time (Target: From 4 hours/week to < 1.5 hours/week)
- **Decision Speed:** Reduce downtime issue identification from 48 hours to same-day
- **Adoption:** Use dashboard in 100% of weekly production reviews

Plant Supervisors

- **Daily Usage:** 80%+ of supervisors check dashboard daily
- **Training Time:** < 2 hours to proficiency
- **Satisfaction:** Net Promoter Score > 7/10

Quality Assurance Manager

- **Data Access:** 100% of requested historical data available
- **Analysis Time:** 50% reduction in time to generate quality reports
- **Integration:** Use dashboard data in monthly QA reports

Production Operators

- **Awareness:** 90%+ understand dashboard purpose
- **Trust:** 75%+ confidence in data accuracy
- **Morale:** No negative impact on employee satisfaction scores

IT Department

- **Support Tickets:** < 5 tickets per month after stabilization
- **System Stability:** 99.5% uptime
- **Security:** Zero security incidents

8. Stakeholder Feedback Mechanisms

Continuous Feedback Channels

- **Suggestion Box:** Anonymous digital submission form
- **Monthly Surveys:** Structured feedback on usability and value
- **Office Hours:** Weekly open sessions for questions
- **User Forum:** Internal collaboration space for tips and issues

Formal Review Points

- **30-Day Review:** Comprehensive feedback from all stakeholder groups
- **90-Day Assessment:** Measure against success criteria
- **Annual Strategic Review:** Roadmap planning with Production Manager

9. Change Champions

Identified Champions

Primary Champion: Senior Plant Supervisor (15 years experience)

Role: Peer advocate, early adopter, training assistant

Responsibilities:

- Demonstrate value to peers
- Provide feedback from floor perspective
- Assist with training sessions
- Address concerns from operators

Secondary Champion: QA Lead Analyst

Role: Data quality advocate

Responsibilities:

- Validate data accuracy
- Promote analytical use cases
- Share success stories

10. Lessons Learned & Best Practices

From Similar Projects

1. **Early Engagement:** Involve end users from day one to build ownership
2. **Quick Wins:** Deliver incremental value to build momentum
3. **Transparent Communication:** Address concerns proactively
4. **Comprehensive Training:** Invest time in user enablement
5. **Continuous Improvement:** Treat dashboard as living product

Application to This Project

- ✓ Conducted pre-project stakeholder interviews
- ✓ Delivered Overview page first for immediate value
- ✓ Planned phased rollout with Batch Detail in Phase 2

- ✓ Documented training materials and user guides
- ☐ Established feedback mechanisms for ongoing refinement

Conclusion

Effective stakeholder management is critical to the success of the Manufacturing Downtime Analysis dashboard. By understanding the needs, concerns, and influence levels of each stakeholder group, we can tailor our engagement approach, address resistance proactively, and ensure the delivered solution meets business objectives while gaining strong user adoption.